Customer Churn Prediction using Machine Learning: A Comparative Study of Decision Tree, Random Forest, and XGBoost

Abstract

Customer churn is one of the major challenges faced by subscription-based businesses, especially in the telecommunications industry. Predicting customers who are likely to discontinue a service allows companies to implement proactive retention strategies and reduce revenue loss. This project presents an end-to-end machine learning solution for telecom customer churn prediction using the IBM Telco Customer Churn dataset. The workflow includes data preprocessing, exploratory data analysis (EDA), feature engineering, model development, and comparative evaluation of three supervised learning algorithms: Decision Tree, Random Forest, and XGBoost. The models were assessed using Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix, and Feature Importance. Experimental results indicate that XGBoost achieved the best balance between Precision and Recall, making it the preferred model for customer churn prediction.

1. Introduction

Customer retention has become increasingly important in highly competitive industries such as telecommunications. Acquiring new customers is often more expensive than retaining existing ones, making churn prediction an essential business application of machine learning. By analyzing customer demographics, subscription details, billing information, and service usage, predictive models can identify customers at high risk of leaving.

The objective of this project is to develop an accurate customer churn prediction model while comparing the performance of multiple machine learning algorithms to identify the most effective approach.

2. Dataset Description

The project utilizes the **IBM Telco Customer Churn Dataset**, which contains demographic information, account details, subscribed services, billing information, and customer churn status.

Dataset Summary

- **Dataset:** IBM Telco Customer Churn
- **Total Records:** 7043
- **Target Variable:** Churn (Yes/No)
- **Problem Type:** Binary Classification

During preprocessing, non-predictive columns such as Customer ID, Churn Score, Churn Value, Churn Reason, Latitude, Longitude, and Zip Code were removed. The **Total Charges** feature was converted from object to numeric format, and missing values were handled appropriately before model training.

3. Methodology

The project followed a structured machine learning pipeline consisting of the following stages:

Data Preprocessing

- Removed irrelevant and redundant features.

- Converted categorical variables using One-Hot Encoding.
- Converted Total Charges to numeric format.
- Handled missing values.
- Split the dataset into training and testing sets (80:20).

Exploratory Data Analysis (EDA)

EDA was performed to understand customer behavior and identify important trends affecting churn.

Key observations include:

- Customers with month-to-month contracts showed the highest churn rate.
- Customers with shorter tenure were more likely to discontinue the service.
- Higher monthly charges were associated with increased churn probability.
- Fiber optic internet users exhibited relatively higher churn.
- Customers paying through electronic checks showed comparatively higher churn rates.

4. Machine Learning Models

Three supervised classification algorithms were implemented and compared.

Decision Tree

A tree-based classification model that recursively partitions the feature space based on decision rules. It provides high interpretability but is prone to overfitting.

Random Forest

An ensemble learning algorithm that combines multiple independently trained decision trees through majority voting, improving prediction accuracy and reducing overfitting.

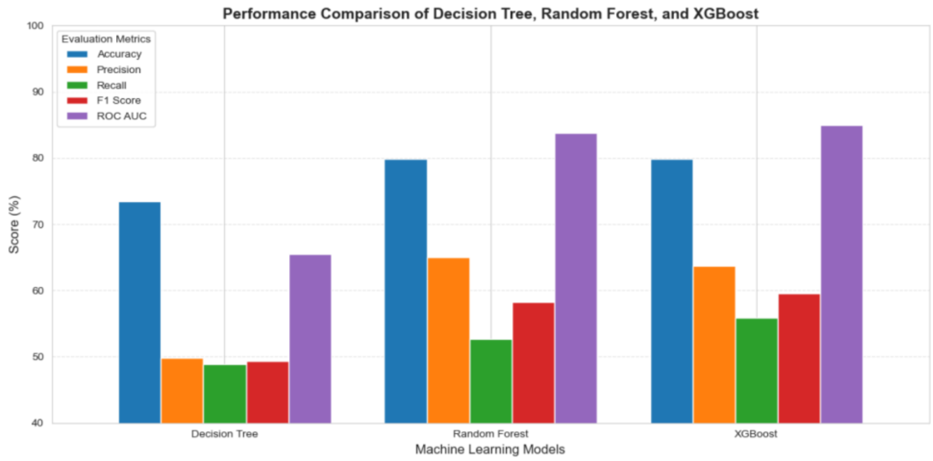
XGBoost

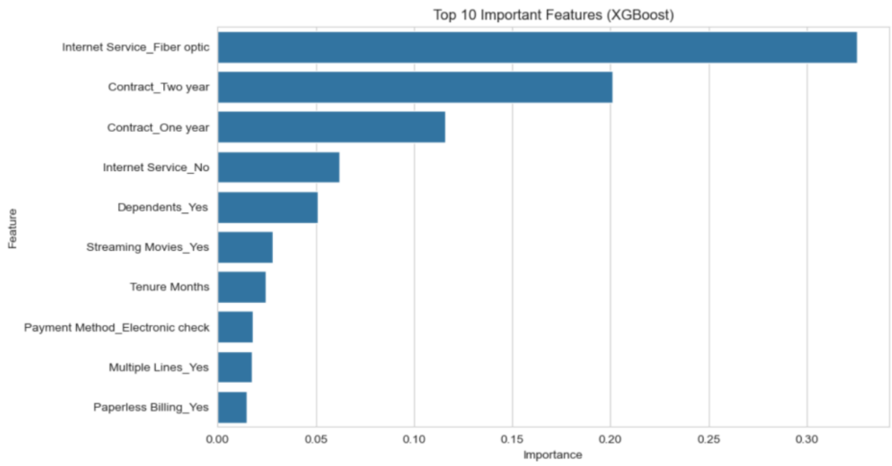
A gradient boosting algorithm where trees are built sequentially, with each new tree learning from the errors of previously constructed trees. This approach improves predictive performance by minimizing residual errors during training.

5. Model Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Decision Tree	**73.39%**	**49.86%**	**48.93%**	**49.39%**	**65.53%**
Random Forest	**79.91%**	**65.23%**	**52.67%**	**58.40%**	**≈84–85%**
XGBoost	**79.84%**	**63.72%**	**55.88%**	**59.54%**	**84.96%**

Although Random Forest achieved the highest classification accuracy, XGBoost demonstrated better Recall and F1-Score, making it more effective at identifying customers likely to churn. Since churn prediction prioritizes identifying at-risk customers, XGBoost was selected as the final model.





6. Key Findings

Feature importance analysis identified the following variables as the most influential predictors of customer churn:

- Tenure
- Total Charges
- Monthly Charges
- Contract Type
- Internet Service
- Payment Method

The analysis indicates that customers with shorter tenure, higher monthly charges, and month-to-month contracts are significantly more likely to discontinue their telecom services.

7. Conclusion

This project successfully developed and evaluated an end-to-end machine learning pipeline for telecom customer churn prediction. Comprehensive data preprocessing, exploratory data analysis, feature engineering, and comparative evaluation of Decision Tree, Random Forest, and XGBoost classifiers were performed. Among the evaluated models, XGBoost achieved the best overall balance between Precision, Recall, and F1-Score, making it the preferred model for predicting customer churn. The project demonstrates the practical application of machine learning techniques for customer retention and business decision support.

8. Future Scope

The performance of the proposed solution can be further enhanced through:

- Hyperparameter tuning using Grid Search or Random Search.
- Handling class imbalance using SMOTE or other resampling techniques.
- Optimizing the classification threshold to improve Recall.
- Deploying the trained model as an interactive web application using Streamlit or Flask.
- Incorporating explainable AI techniques such as SHAP to improve model interpretability.

References

1. IBM Telco Customer Churn Dataset.
2. Scikit-learn Documentation.
3. XGBoost Documentation.
4. Pandas Documentation.
5. Matplotlib and Seaborn Documentation.